

Sarah Ali

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🌐 [Sarah Ali](#)

WHO AM I?

Electrical Engineering graduate from **NUST** with combined experience in embedded systems and product/design engineering. Hands-on background in embedded firmware development, UAV/drone prototyping, and hardware-software integration using **ESP32/STM32** platforms, sensors, and **PID/control** systems. Basic **PCB prototyping**, embedded debugging, and system testing, alongside strong **CAD capabilities in Fusion 360 and SolidWorks** with **3D printing for rapid** prototyping. Experienced in developing end-to-end engineering solutions from concept design to functional embedded hardware implementation through academic and project work, including UAV systems

RELEVANT EMBEDDED | FLIGHT ENGINEER PROJECTS

- **Flight Engineer | DRONEXT Lab (SEECs)**
- **Final Year Project**
 - Spearheaded the development of a **courier drone**, integrating flight controllers, navigation algorithms, and embedded electronics to achieve autonomous delivery capabilities.
- **Bicopter Flight Controller (ESP32, PID Control)**

Designed, tuned, and executed a bi-copter flight controller using ESP32. Achieved stable flight, disturbance rejection, and precise PID gains for real-time control.
- **Self-Balancing Robot (ESP32, PID)**

Engineered a near-complete PID-controlled balancing robot using ESP32—demonstrating real-time feedback integration and motion control.
- **Drone Auxiliary Control via ESP32 + GSM**

In my FYP, I designed the whole drone(fully original designed airframe) and integrated an ESP32 GSM module as a secondary controller on Drone, enabling GSM-based OTP communication with users—enhancing drone interaction and control.

EXPERIENCE

Intern – DroNexas SEECs

(June – September 2025)

- Designed 3 UAVs in Fusion 360 and 3D printed prototypes.
- Contributed to the **Tethered Drone System** showcased on 14th August Parade.
- Gained hands-on experience in **flight controllers, ESCs, and 3D design**.

Quality Assurance Engineer – 9t5 Pvt. Ltd., Lahore

(June – September 2024)

- Designed and executed test plans and cases for web and mobile apps.
- Performed **manual and automated testing** (functional, regression, performance).
- Collaborated with developers to resolve bugs, improving release quality.

EDUCATION

Degree	Year	Institute	Grades
BS. Electrical Engineering	2026	NUST, Islamabad	CGPA: 3.3

PROJECTS

PID Tuned Flight Controller of Bicopter –ESP32	Bluetooth Controlled Circuit Board -Arduino
Drone Auxiliary Control-ESP32 + GSM	Temp Controlled Cooling System
Tethered Drone System (Quadcopter)	Voltage Overcharge Protection
Arduino based Bluetooth Control Car	Wireless Charger – using PCB
Digital Voting Machine- FPGA	Digital Multimeter- Arduino
Concordia1 3D speakers – Fusion 360	Designed Drone Testing Net – Fusion 360
Courier Drone – Fusion 360	Package Drone – Fusion 360

TECHNICAL SKILLS & EMBEDDED COMPETENCIES

- **Design & CAD:** Fusion 360, SolidWorks, AutoCad, 3D printing workflows.
 - **UAV / Avionics Systems:** Drone Flight Controllers (ESP32, Pixhawk/Ardupilot), UAV electronics, ROS2
 - **Connectivity:** Wi-Fi, GSM, secure communications
 - **Microcontrollers:** Extensive experience with Arduino **ESP32**, familiar with STM32, FPGA, Raspberry Pi architectures
 - **Control Systems:** PID tuning, real-time loop control, sensor integration
 - **Development Tools:** Familiar with **VS Code**, **Arduino IDE**, **STM32Cube**, **Quartus** etc.
 - **Hardware Design:** PCB prototyping, sensor interfacing, power optimization
 - **Embedded Software:** C/C++ programming, real-time debugging, Git version control
 - **Simulation & Analysis:** MATLAB/Simulink, ANSYS MoterCad (basic)
 - **Other tools:** AI tools, Microsoft Word, PowerPoint, Excel
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